

Run Time Data: 2 Large Primes + Multiple Messages
BIT SIZE: 16

large prime (p)	large prime (q)	m1 (1 to q/3)	m1 attack run time	m2 (q/3 to 2q/3)	m2 attack run time	m3 (2q/3 to q)	m3 attack run time	phi(N)	N (p x q)	d	Weiner's Attack Bound	Is d < 1/3 * N ^{1/4}	e
45259	57989	2062	0.001488016	28041	0.006505020	56224	0.01255383	2620347864	2620450841	524069637	75.41760724	No	5
37907	61463	4673	0.000664234	39879	0.005321195	51428	0.007724047	2329778572	2329877941	1553185715	73.23387892	No	3
42061	54787	8232	0.002231121	24500	0.006916046	47642	0.01396203	2304299160	2304369007	967556783	73.03281218	No	7
43577	53003	11172	0.001591206	20182	0.003676792	50420	0.00873208	2309615152	2309711731	1539743435	73.07489337	No	3
48017	62311	3661	0.001064062	26891	0.008612156	41828	0.01262106	2991876960	2991987287	1282232983	77.95943469	No	7
48871	50263	14824	0.004978895	33835	0.011723042	43401	0.015100002	2896133940	2896242073	789854711	77.32812144	No	11
45049	50497	15100	0.003125906	23042	0.00494194	52600	0.011536837	2733722208	2733827653	1640233325	76.22045348	No	5
39239	60919	5387	0.001348257	26262	0.005623817	57104	0.012686253	2390300484	2390400641	478060097	73.70490987	No	5

BIT SIZE: 32

large prime (p)	large prime (q)	m1 (1 to q/3)	m1 attack run time	m2 (q/3 to 2q/3)	m2 attack run time	m3 (2q/3 to q)	m3 attack run time	phi(N)	N (p x q)	d	Weiner's Attack Bound	Is d < 1/3 * N ^{1/4}	e
2470285981	2808859103	510962362	211.8311031	1230201645	522.5250661	1905085283	831.5198958	6.93869E+18	6.93869E+18	1.89237E+18	17107.95276	No	11
2253280847	3878192183	248915626	38.36964011	2291364475	405.5347352	3086481498	573.136761	8.73866E+18	8.73866E+18	5.82577E+18	18123.4097	No	3
2226149353	3880406611	558848542	253.751735	2055544117	972.2417172	3226602328	1547.150009	8.63836E+18	8.63836E+18	1.57061E+18	18071.18487	No	11
3453008519	4108358041	1088075549	417.2864687	1695441068	666.2414372	4094241772	3466.619703	1.41862E+19	1.41862E+19	4.0532E+18	20457.15468	No	7
2493753169	3611550347	371690942	91.88447571	1583958784	409.2716439	2794803771	770.2752922	9.00632E+18	9.00632E+18	5.40379E+18	16260.62046	No	5
2284075831	3256974103	426824454	137.6898562	1086815833	354.2409486	2857048322	2787.228112	7.43712E+18	7.43712E+18	6.37467E+18	17407.24111	No	7
2598290723	2729079183	367276310	85.85974193	1000935663	688.622972	2078413452	563.931725	7.09086E+18	7.09086E+18	1.41817E+18	17209.90699	No	5
2452317611	3458585041	873661009	281.4533482	1698807830	570.9048541	2429520552	831.5770872	8.48155E+18	8.48155E+18	7.2699E+18	17998.60722	No	7

Run Time Data: 1 Large Prime, 1 Small Prime + Multiple Messages
BIT SIZE: 16

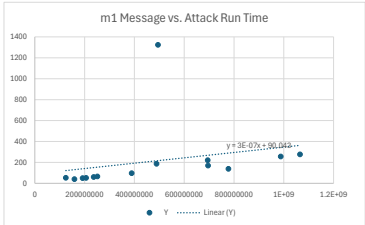
small prime (p)	large prime (q)	m1 (1 to q/3)	m1 attack run time	m2 (q/3 to 2q/3)	m2 attack run time	m3 (2q/3 to q)	m3 attack run time	phi(N)	N (p x q)	d	Weiner's Attack Bound	Is d < 1/3 * N ^{1/4}	e
3	64483	11509	0.001521826	38266	0.005178213	44463	0.005998943	4473710256	193449	25793	6.99696521	No	5
5	52177	7873	0.001090557	28905	0.003573996	45057	0.006081104	15032186232	269885	41741	7.533399911	No	5
7	43597	7328	0.000960112	28349	0.003810883	41995	0.005667686	14676306696	305179	209261	7.834610435	No	5
11	36013	11217	0.001721144	16059	0.002050302	28307	0.004423857	21476953920	396143	205783	8.362608044	No	7
13	58907	7256	0.000964165	31212	0.004075766	56876	0.007534981	32844941784	765791	282749	9.860667388	No	5
43	62213	18606	0.002438784	37378	0.004957914	43598	0.005785227	1.70905E+11	2675159	522581	13.48081489	No	5
71	42557	4720	0.000509977	23258	0.002494812	41433	0.004491806	2.29662E+11	3021547	1869547	13.89748083	No	11
97	33797	7001	0.000901699	13710	0.001755953	23079	0.003057718	4.04566E+11	3278309	2595533	14.18375486	No	5
131	56569	1843	0.000280857	27607	0.004510131	46566	0.007044077	4.69882E+11	7410539	5252743	17.39166633	No	3
157	61403	9502	0.00123016	36083	0.00463891	54791	0.00722003	3.96737E+11	9640271	3831485	18.57381245	No	5
383	61651	6358	0.000957012	26167	0.004015923	58215	0.008894205	9.24341E+11	23612333	16821643	23.23611248	No	5
727	58921	11105	0.001749039	25530	0.004065037	44991	0.006800235	2.73909E+12	42835567	24443383	26.96682863	No	5
967	48163	9289	0.001250029	27219	0.003479958	34563	0.00447011	2.40911E+12	46573621	18609797	27.53681751	No	5
1555	60449	4192	0.000549793	29578	0.003831148	55140	0.007237911	4.4405E+12	93998195	37574477	32.82151501	No	5

BIT SIZE: 32

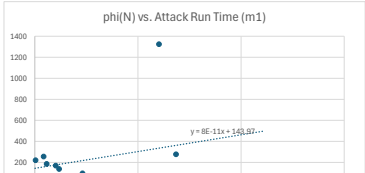
small prime (p)	large prime (q)	m1 (1 to q/3)	m1 attack run time	m2 (q/3 to 2q/3)	m2 attack run time	m3 (2q/3 to q)	m3 attack run time	phi(N)	N (p x q)	d	Weiner's Attack Bound	Is d < 1/3 * N ^{1/4}	e
3	238685129	736706309	60.84448981	1233375000	328.6299419	2213791011	608.717612	4473710256	6710565387	357968205	95.4044261	No	5
5	3758046559	251390067	65.75115108	1588920607	2118.757551	3255821355	4849.900221	15032186232	18790232795	601287493	123.4132489	No	5
7	2446051117	694644410	221.4196022	848859086	264.829694	2198045343	770.6508699	14676306696	17122357819	11741045357	120.5784335	No	5
11	2147695393	124692973	53.5274222	1262603711	554.3923337	1451546666	4406.175386	21476953920	23624649323	9204408823	130.6832986	No	7
13	2737078483	191915631	49.97334909	1583418235	428.8001459	2094751590	582.0483112	32844941784	35582020279	6568988357	144.7726073	No	5
43	4069173223	988031875	256.3360982	2159196601	588.0673249	3097081376	865.4431298	1.70905E+11	1.74974E+11	34181050065	215.5869179	No	5
71	3280890787	489123687	186.458998	2021228249	801.2938941	3260318777	1325.953294	2.29662E+11	2.32943E+11	20878395911	231.5747884	No	11
97	4214227013	699950190	170.0725121	1908224415	483.0854662	3195468788	835.4196309	4.04566E+11	4.0678E+11	1.61826E+11	266.5331061	No	5
131	361448027	777710672	138.2504361	2382462325	461.099638	288648767	569.7828832	4.69855E+11	7.34897E+11	3.13255E+11	276.5082073	No	5
157	2543186753	158318117	39.94909787	1295160910	1331.1481762	1889187755	497.5647087	3.96737E+11	3.9928E+11	1.58695E+11	264.9706248	No	5
383	2419739983	389173106	96.57160091	1042602666	259.615571	2249930830	588.1314189	9.24341E+11	9.2676E+11	1.84868E+11	327.0548652	No	5
727	3772853657	1065003063	277.3569911	1497325099	398.6155059	2516845420	701.188971	2.73909E+12	2.74286E+12	2.19127E+12	428.9728813	No	5
967	2493907349	494534067	1324.73275	878957542	227.3499939	2049956228	560.4794402	2.40911E+12	2.41161E+12	1.44547E+12	415.3894312	No	5
1555	2857466459	209952671	52.48355699	1606031205	427.5955174	1937755595	537.2894161	4.4405E+12	4.44336E+12	1.7762E+12	483.9568362	No	5

Graphs Based on Data
1 LARGE PRIME + 1 SMALL PRIME W/ 32-BIT NUMBERS

Message (m1)	Attack Run Time
X	Y
124692973	53.52744222
159318117	39.94909787
191915631	49.97334909
209952671	52.48355699
226706309	60.84448981
251390067	65.75115108
389173106	96.57160091
489123687	186.458998
494534067	1324.73275
694644410	221.4196022
695950190	170.0725121
777710672	138.2504361
988031875	256.3360982
1065003063	277.3569911



phi(N)	Attack Run Time (m1)
X	Y
4473710256	60.84448981
14676306696	221.4196022
15032186232	65.75115108
21476953920	53.52744222
32844941784	49.97334909
1.70905E+11	256.3360982
2.29662E+11	186.458998
3.96737E+11	39.94909787
4.04566E+11	170.0725121
4.69882E+11	138.2504361



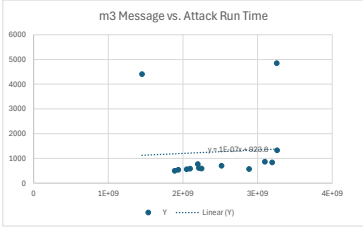
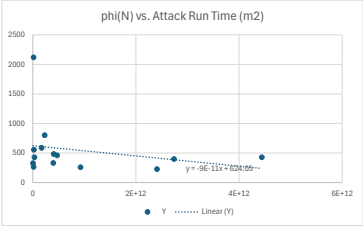
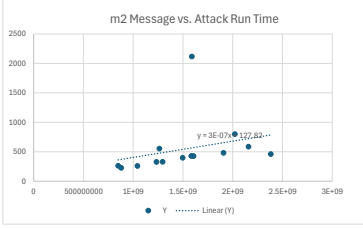
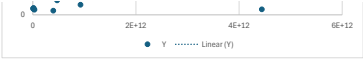
9.24341E+11	96.57160091
2.40911E+12	1324.73275
2.73909E+12	277.3569911
4.4405E+12	52.48355699

Message (m2)	Attack Run Time
X	Y
1233275000	328.6290419
1588992067	2118.757551
848859086	264.829694
1262603711	554.3923337
1583418235	428.8001459
2159196601	588.9673249
2021229249	801.3938941
1908224415	483.0654662
2382462325	461.0898828
1295160910	331.1481762
1042602666	259.615571
1497325099	398.6155059
878957542	227.3499939
1806031205	427.5955174

phi(N)	Attack Run Time (m2)
X	Y
4473710256	328.6290419
14676306996	264.829694
15032186232	2118.757551
21476953920	554.3923337
32844941784	428.8001459
1.70905E+11	588.9673249
2.29692E+11	801.3938941
3.96737E+11	331.1481762
4.04566E+11	483.0654662
4.69882E+11	461.0898828
9.24341E+11	259.615571
2.40911E+12	227.3499939
2.73909E+12	398.6155059
4.4405E+12	427.5955174

Message (m3)	Attack Run Time
X	Y
2213791011	608.7171612
3255821355	4849.900221
2198045343	770.6508689
1451546666	4406.175386
2094751590	582.0483112
3097081376	865.4431298
3260318777	1325.952914
3195468788	835.4106309
2886486747	569.7828832
1889187755	497.5647087
2249930830	588.1314189
2516845420	701.188971
2049956228	560.4794402
1937745595	537.2894161

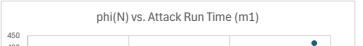
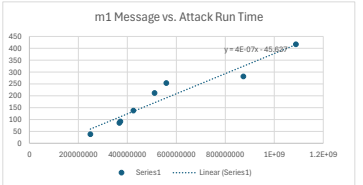
phi(N)	Attack Run Time (m3)
X	Y
4473710256	608.7171612
14676306996	770.6508689
15032186232	4849.900221
21476953920	4406.175386
32844941784	582.0483112
1.70905E+11	865.4431298
2.29692E+11	1325.952914
3.96737E+11	497.5647087
4.04566E+11	835.4106309
4.69882E+11	569.7828832
9.24341E+11	588.1314189
2.40911E+12	560.4794402
2.73909E+12	701.188971
4.4405E+12	537.2894161



2 LARGE PRIMES W/ 32-BIT NUMBERS

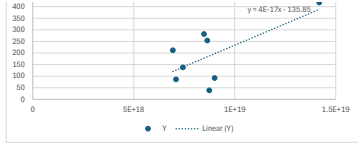
Message (m1)	Attack Run Time
X	Y
248915626	38.36964011
367276310	85.85974193
371690942	91.88447571
424624454	137.6898582
510962362	211.8311031
558848542	253.751735
873661009	281.4533482
1088075549	417.2864687

phi(N)	Attack Run Time (m1)
--------	----------------------

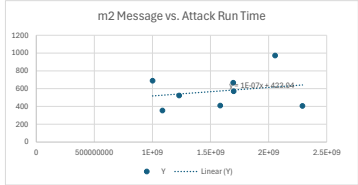


Message Size	Brute Force Attack
X	Y
248915626	38.36964011
367276310	85.85974193
371690942	91.88447571
424624454	137.6898582
510962362	211.8311031
558848542	253.751735
873661009	281.4533482
1088075549	417.2864687
100935663	688.622972
1096615833	354.204946
1230201645	522.5250661
1583958784	409.2716439
1695441068	666.2414372
1698807830	570.9048541
2055544117	972.2471712

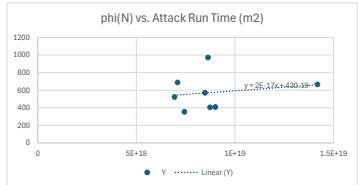
X	Y
6.93869E+18	211.8311031
7.09086E+18	85.8974193
7.43712E+18	137.6898582
8.46155E+18	261.4533482
8.63836E+18	253.751735
8.73866E+18	36.36964011
9.00632E+18	91.88447571
1.41862E+19	417.2864687



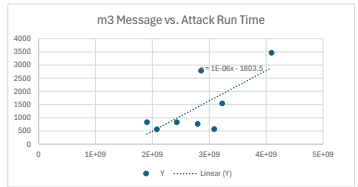
Message (m2)	Attack Run Time
X	Y
1000935663	688.622972
1086615833	354.204946
1230201645	522.5250661
1583958784	409.2716439
1695441068	666.2414372
1698807930	570.9048541
2055544117	972.2471712
2291364475	405.5347352



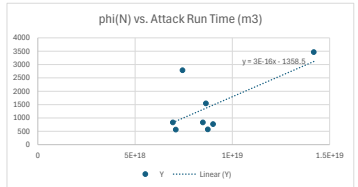
phi(N)	Attack Run Time (m2)
X	Y
6.93869E+18	522.5250661
7.09086E+18	688.622972
7.43712E+18	354.204946
8.46155E+18	570.9048541
8.63836E+18	972.2471712
8.73866E+18	405.5347352
9.00632E+18	409.2716439
1.41862E+19	666.2414372



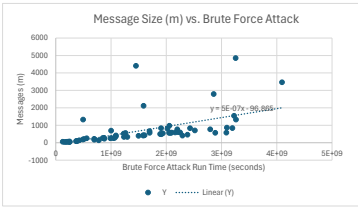
Message (m3)	Attack Run Time
X	Y
1905085283	831.5198958
2078413452	563.931725
2429552052	831.5770872
2794903771	770.2752922
2857049322	2787.228112
3086481498	573.136761
3226602328	1547.150009
4094241772	3466.619703



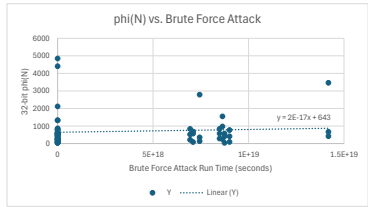
phi(N)	Attack Run Time (m3)
X	Y
6.93869E+18	831.5198958
7.09086E+18	563.931725
7.43712E+18	2787.228112
8.46155E+18	831.5770872
8.63836E+18	1547.150009
8.73866E+18	573.136761
9.00632E+18	770.2752922
1.41862E+19	3466.610703



2291364475	405.5347352
1905085283	831.5198958
2078413452	563.931725
2429552052	831.5770872
2794903771	770.2752922
2857049322	2787.228112
3086481498	573.136761
3226602328	1547.150009
4094241772	3466.619703
2213791011	608.7171612
3255821355	4849.900221
2198045343	770.6506889
1451546666	4406.175386
2094751590	562.0463112
3097081376	865.4431296
3260318777	1325.952014
3195468788	835.4106309
2886486747	569.7828832
1889187755	497.5647087
2249930830	588.1314189
2516845420	701.188971
2049956228	560.4794402
1937745595	537.2894161
1233275000	326.6290419
1588992067	2118.757551
848859086	264.829694
1262603711	554.3923337
1583418235	428.8001459
2159196601	588.0673249
2021228249	801.2938941
1060224415	463.0654652
2382462325	461.0896828
1295160910	331.1481762
1042602666	259.615571
1497325099	398.6155059
878957542	227.3499939
1606031205	427.5955174
124692973	53.52744222
159318117	39.94090787
191915631	49.97334909
205952671	52.48355699
236706309	60.84448981
251390067	65.75115108
389173106	96.57160091
489123687	186.458998
494534067	1324.73275
694644410	221.4196022
696959190	170.0725121
777710672	138.2504361
988031875	256.3360982
1065003063	277.3569911



phi(N)	Attack Run Time (all messages)
X	Y
4473710256	60.84448981
14676306696	221.4196022
15032186232	65.75115108
21476953920	53.52744222
32844941784	49.97334909
1.70905E+11	256.3360982
2.20682E+11	186.458998
3.96737E+11	39.94090787
4.04566E+11	170.0725121
4.69882E+11	138.2504361
9.24341E+11	96.57160091
2.40911E+12	1324.73275
2.73909E+12	277.3569911
4.4405E+12	52.48355699
4473710256	328.6290419
14676306696	264.829694
15032186232	2118.757551
21476953920	554.3923337
32844941784	428.8001459



1.70905E+11 2.29662E+11 3.96737E+11 4.04566E+11 4.69882E+11 9.24341E+11 2.40911E+12 2.73909E+12 4.4405E+12	588.0673249 801.2938941 331.1481762 483.0654662 461.0896828 259.615571 227.3499939 398.6155059 427.5955174
4473710256 14676306696 15032186232 21476953920 32844941784 1.70905E+11 2.29662E+11 3.96737E+11 4.04566E+11 4.69882E+11 9.24341E+11 2.40911E+12 2.73909E+12 4.4405E+12	608.7171612 770.6506889 4849.000221 4406.175386 582.0483112 865.4431298 1325.952914 497.5647087 835.4106309 569.7828832 588.1314189 560.4794402 701.188971 537.3994161
6.93866E+18 7.09086E+18 7.43712E+18 8.48155E+18 8.63836E+18 8.73866E+18 9.00632E+18 1.41962E+19	211.8311031 85.85974193 137.6898582 281.4533482 253.751735 38.39994011 91.88447571 417.2864687
6.93866E+18 7.09086E+18 7.43712E+18 8.48155E+18 8.63836E+18 8.73866E+18 9.00632E+18 1.41962E+19	522.5250661 688.622972 354.204946 570.9048541 972.2471712 405.5347352 409.2716439 666.2414372
6.93866E+18 7.09086E+18 7.43712E+18 8.48155E+18 8.63836E+18 8.73866E+18 9.00632E+18 1.41962E+19	831.5198958 563.931725 2787.228112 831.5770872 1547.150009 573.136761 770.2752922 3496.619703